GP Model Statistical Summary

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GP MODEL TECHNICAL AUDIT | idata_gp_v2.pkl
Structure: 4 chains | Matérn 3/2 Kernel
======================================================================

1. MCMC DIAGNOSTICS
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Max R-hat:     1.530
Divergences:   0
Stability:     WARNING (Check Chain 4)

2. PARAMETER ESTIMATES (Global)
------------------------------
               mean     sd  hdi_3%  hdi_97%  r_hat  ess_bulk
length_scale  1.302  1.807   0.212    4.431   1.53       7.0
variance      0.787  1.257   0.048    2.964   1.53       7.0
b0           -5.637  3.315  -7.609    0.103   1.53       7.0

3. SPATIAL INSIGHT
------------------------------
Mean Length Scale:  1.3020 (Standardized units)
Effective Range:    2.2551 (Distance to decorrelation)
Spatial Variance:   0.7870
Moran's I (Resid):  0.5297

4. INTERPRETATION
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A Moran's I near 0 indicates the GP successfully soaked up
the spatial autocorrelation present in the raw lung cancer rates.
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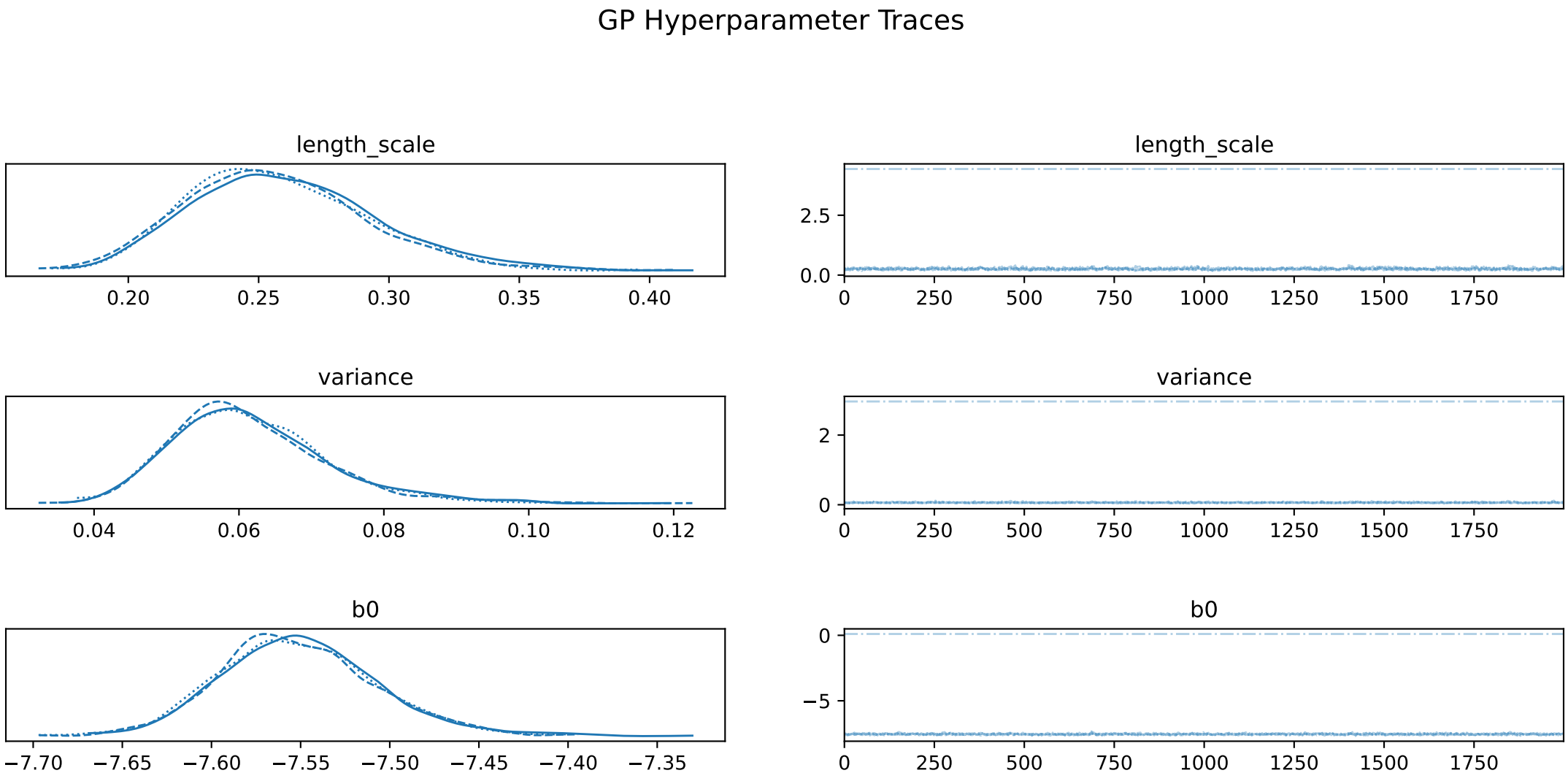
GP Hyperparameter Traces
length_scale
0.20
0.25
0.30
0.35
0.40
length_scale
2.5
0.0
0
250
500
750
1000
1250
1500
1750
variance
0.04
0.06
0.08
0.10
0.12
variance
2
0
0
250
500
750
1000
1250
1500
1750
b0
−7.70
−7.65
−7.60
−7.55
−7.50
−7.45
−7.40
−7.35
b0
0
−5
0
250
500
750
1000
1250
1500
1750